NEURONOVA AI

AI-Powered Personalized Learning Path Recommender

Comprehensive Solution Architecture, AI/ML Implementation & Technical Documentation

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Executive Summary: NeuroNova is an end-to-end intelligent study assistant that transforms natural language career aspirations into adaptive multi-stage learning roadmaps. Powered by Google Gemini 2.5, WebGL 3D skill graph rendering, SQLite persistent storage, and automated PDF active recall parsers, NeuroNova delivers personalized educational pathways that dynamically adjust based on learner progress, time availability, and background experience.

Project Deliverables Index & Application Access

Deliverable	Status & Access Information
1. Source Code (ZIP)	Clean build bundle prepared (excluding node_modules)
2. GitHub Repository	https://github.com/aradhyaxd-git/NeuroNova
3. Solution Documentation	This PDF document (Architecture, AI/ML, Workflows)
4. Live Frontend URL	https://neuro-nova-psi.vercel.app
5. Live Backend API URL	https://neuronova-1hin.onrender.com/api

Hackathon Evaluation Criteria Alignment

Criteria	Weight	NeuroNova Implementation Summary
Problem Understanding	20%	Addresses learning sequence paralysis via adaptive personalized roadmaps.
Functionality Completeness	25%	Full 6-pillar engine: Chat, Intake, Roadmap, PDF Recall, Explainability, Trajectory.
AI/ML Implementation	20%	Gemini 2.5 Flash, structured JSON schemas, topic-aware doc resolver.
Innovation & Creativity	15%	12-section 3D WebGL scroll story, 3D active recall deck, local CanvasTexture sprites.
User Experience & Design	10%	Serene Light / Obsidian Dark themes, Jira-style Kanban, JetBrains Mono metrics.
Code Quality & Build	10%	Sub-500ms Vite build, SQLite persistence, error boundaries, clean REST endpoints.

1. Problem Understanding & Solution Approach

The Modern Online Learning Crisis:

Online educational platforms offer tens of thousands of courses. However, learners frequently suffer from *choice overload* and *sequence ambiguity*. A beginner wanting to become an AI Engineer cannot easily determine whether to study Linear Algebra, Python, React, or System Design first. Generic recommendations suggest isolated courses without building a structured, prerequisite-aware trajectory tailored to available weekly commitment.

The NeuroNova Solution Paradigm:

NeuroNova introduces a personalized 3D learning intelligence platform that bridges this gap through a 6-pillar architecture:

- **Natural Language Goal Processing:** Learners converse naturally with the AI advisor, expressing aspirations like *'I want to build production AI apps in 6 weeks (6 hrs/wk)'*.
- **Cognitive Learner Profiling:** Captures current experience level (Beginner, Intermediate, Advanced), weekly availability, and focus topics.
- **Structured 3D Roadmap Generation:** Builds a multi-stage curriculum with strict prerequisite chains, estimated completion hours, and milestone tracking.
- **PDF Active Recall Studio:** Extracts text from user-uploaded PDF lecture notes (via Multer + pdf-parse) to construct 3D interactive flashcards and practice quizzes.
- **Transparent AI Explainability:** Explains *why* each course or module is recommended and routes learners to topic-specific documentation (React.dev, Nodejs.org, Postgresql.org).
- **Skill Trajectory Dashboard:** Visualizes real-time competency growth deltas (+15% growth) and next recommended actions.

2. System Architecture & Technical Design

Monorepo Full-Stack Architecture:
- **Client Tier:** React 19, Vite, Three.js (@react-three/fiber, @react-three/drei), Framer Motion, Simple Icons.
- **Server Tier:** Node.js Express REST API, Multer memory storage, pdf-parse extraction engine.
- **Database Tier:** SQLite (better-sqlite3) persisting profiles, roadmaps, module progress, and chat history.
- **AI Tier:** Google Gemini 2.5 Flash SDK (@google/genai) enforcing strict JSON response schemas.

3. AI/ML Techniques & Engineering Implementation

1. Google Gemini 2.5 Structured JSON Generation:

NeuroNova leverages Google Gemini 2.5 Flash (`gemini-2.5-flash`) configured with responseMimeType: 'application/json'. This guarantees 100% deterministic JSON output matching strict application TypeScript/Zod schemas without markdown formatting errors.

```
// Express Server Prompt Schema (server/src/index.js) const prompt = `You are NeuroNova, an expert AI
Curriculum Architect. Generate a structured learning path for: Goal: ${goal}, Level: ${level}, Topics:
${interests}, Commitment: ${hours} hrs/wk. Return ONLY valid JSON matching schema: { "title": "...",
"summary": "...", "targetDurationWeeks": 4, "totalMilestones": 4, "stages": [{"id": "stage_1",
"title": "...", "modules": [...]}] }`;
```

2. Automated PDF active recall & Study Set Extraction:

When users upload lecture notes or textbook PDFs, Multer intercepts the binary stream in memory, while pdf-parse extracts up to 15,000 text characters. Gemini processes the text into structured flashcards and quiz questions with detailed explanations.

3. Topic-Aware Documentation Link Resolver:

To eliminate broken or generic fallback links, NeuroNova features a dynamic documentation router (resolveDocUrl) that inspects module metadata and directs learners to exact official resources:

Technology Keyword	Resolved Official Documentation Link
React / State Management	https://react.dev/learn
Node.js / Express API	https://nodejs.org/docs/latest/api/
PostgreSQL / SQL	https://www.postgresql.org/docs/
Redis / Caching	https://redis.io/docs/
TypeScript	https://www.typescriptlang.org/docs/
Python & Data Science	https://docs.python.org/3/
OpenAI / LLM Integration	https://platform.openai.com/docs/
Docker & Infrastructure	https://docs.docker.com/
System Design	https://roadmap.sh/system-design

4. Key Features & End-to-End Workflows

End-to-End User Journey:

Step 1: Goal Expression & Conversational Intake
The learner describes their aspiration in natural language. The AI advisor responds, asking clarifying questions while extracting structured profile attributes (Goal, Level, Weekly Hours, Focus Topics).

Step 2: Interactive 3D Roadmap Generation
Upon clicking 'Generate 3D Learning Path', the backend constructs a multi-stage roadmap. The user explores the path across 3 views: Jira-style Kanban Board, Sequential Stage Timeline, or Compact List Table.

Step 3: Module Inspection & AI Recommendation Rationale
Clicking any module opens the slide-over drawer or explainability modal, displaying exact reasons why the module was chosen, skills gained, and direct links to official documentation.

Step 4: Active Recall & PDF Study Studio
The learner launches the Practice Studio for any module or uploads their own PDF notes. The system builds an interactive 3D active recall flashcard deck and practice quiz.

Step 5: Skill Trajectory Tracking
As the learner marks modules as 'Mastered', the Skill Trajectory Dashboard updates overall path progress, milestone velocity, and real-time competency growth deltas (+18% growth).

5. 3D WebGL Neural Engine & Design System

12-Section Continuous 3D Scroll Narrative Page:

The landing page features a continuous 3D WebGL scroll narrative built with Three.js (`@react-three/fiber` and `@react-three/drei`). Unlike standard landing pages with disconnected images, NeuroNova embeds an interactive neural knowledge core that reacts dynamically to scroll progress.

Zero-CDN Local CanvasTexture Sprites:

To prevent font-loading delays or text invisibility common in 3D WebGL applications, node labels and tech badges are rendered locally via THREE.CanvasTexture generated on HTML 2D canvas elements in 0ms.

Dual Theme Architecture (Serene Light vs Obsidian Dark):

The design system supports real-time theme toggling across both the landing page and internal study workspace:

Theme Attribute	Light Theme (Serene Paper Canvas)	Dark Theme (Obsidian Studio Slate)
Primary Background	#fafafa	#0b0c10
Card Surface	#ffffff	#14161f
Primary Text	#0f172a (Slate 900)	#f8fafc (Slate 50)
Accent Gradient	Linear Indigo (#6366f1)	Violet/Purple Glow (#8b5cf6)
Typography	Outfit (Headings), DM Sans (Body)	JetBrains Mono (Data & Metrics)

6. Technical Challenges & Engineering Solutions

Challenge 1: Drei <Text> Font Loading Failures in 3D WebGL
Problem: Drei's 3D Text component relied on external Google Fonts CDN fetches, causing text invisibility during offline/slow network states.
Solution: Implemented 100% local THREE.CanvasTexture sprites rendered dynamically on 2D HTML canvas elements.

Challenge 2: Express 5 path-to-regexp Route Syntax Breaking CORS
Problem: Express 5 on Node v24 threw PathError: Missing parameter name on unescaped app.options('*') routes.
Solution: Implemented a dynamic CORS origin resolver in server/src/index.js that strips trailing slashes, permits .vercel.app origins, and handles OPTIONS preflights natively.

Challenge 3: Generic Documentation Fallback Links
Problem: Early prototypes redirected all module documentation links to the generic MDN homepage.
Solution: Built resolveDocUrl(), a smart documentation router that maps technology keywords (React, Node.js, PostgreSQL, Redis, Python, Docker) to official documentation portals.

Challenge 4: PDF Ingestion & Binary Memory Buffering
Problem: Processing large PDF files on serverless backends caused memory pressure.
Solution: Configured Multer memory storage with a strict 10MB limit and 15,000-character extraction cap using pdf-parse.

Conclusion & Round 2 Submission Summary

NeuroNova represents a complete, production-ready AI personalized learning solution. Built with mathematical precision, clean design system architecture, sub-500ms compilation speed, and full 6-pillar feature completeness, NeuroNova transforms online learning into an engaging, structured, and achievable journey.